

Contemporary Pharmacotherapy for Resistant Hypertension: A Systematic Literature Review and Meta-Analysis

Catherine Tremblay, MD; Jean-Luc Moreau, MD
Department of Cardiology, Montreal Heart Institute & McGill University,
Canada

ABSTRACT

Resistant hypertension affects an estimated 10-15% of treated hypertensive populations globally. We systematically reviewed 34 randomized clinical trials comprising 28,450 total participants to evaluate the comparative blood pressure lowering efficacy and aggregate class adverse event rates of third-generation vasodilators. No new individual clinical safety cases are reported.

INTRODUCTION & OBJECTIVES

Resistant hypertension remains an independent risk factor for adverse cardiovascular events, heart failure, and renal disease progression. Multiple pharmaceutical classes, including mineralocorticoid receptor antagonists and novel smooth muscle relaxants, have emerged. This review synthesizes aggregate population-level evidence.

METHODOLOGY

Systematic electronic searches of MEDLINE, Embase, and the Cochrane Central Register of Controlled Trials were executed from January 2015 to October 2025. Studies reporting aggregate clinical outcomes were pooled using random-effects meta-analytic models.

SUMMARY OF FINDINGS

Overall pooled analysis demonstrated a weighted mean reduction in systolic blood pressure of 8.4 mmHg (95% CI: 6.2–10.6). Across all pooled cohorts, aggregate adverse event frequencies for dizziness, headache, and peripheral edema were comparable to placebo controls. No individual patient-level identifiers or novel safety signals were identified in the literature sample.

REGULATORY TRIAGE NOTE

Systematic reviews and aggregate meta-analyses devoid of individual identifiable patient case reports do not constitute ICSR-reportable sources under global pharmacovigilance regulations.

REFERENCES

1. Carey RM, et al. Resistant Hypertension: Detection, Evaluation, and Management. Hypertension. 2020;72:e53-e90.